

YT-SWIN: Context-Adaptive Spectral Fusion for Cross-Modal UAV Remote Sensing

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ABSTRACT Accurate object detection in multimodal remote sensing images remains a significant challenge for autonomous all-weather systems, necessitating adaptive integration of complementary RGB and infrared (IR) data under diverse environmental conditions. Existing fusion approaches often rely on fixed, manually designed strategies or complex multi-expert frameworks, which lack the flexibility to dynamically achieve fusion policies in response to real-time sensor degradation. To address these limitations, we propose YT-SWIN, a novel adaptive fusion framework that synergistically combines the YOLOv10n backbone with a Swin Transformer-based attention mechanism to achieve environment-aware multimodal perception. Our work introduces three core modules: first, Self-Supervised Environment Inference Module that learns to characterize weather and illumination conditions directly from input data, eliminating the reliance on labeled environmental metadata. Second, a Learnable Frequency Decomposition Block that constructs optimized spectral representations through data-adaptive filtering, enhancing multimodal feature representation. Third, a Hierarchical Cross-Modal Attention Network (HCAN) built upon lightweight Swin Transformer blocks dynamically gates and fuses RGB and IR features across multiple spatial and frequency scales, conditioned on the inferred environmental context. Additionally, an adaptive Channel Pruning Strategy integrated into the multi-scale feature pyramid maintains real-time inference speeds without compromising detection accuracy. Extensive experiments demonstrate that YT-SWIN achieves state-of-the-art performance, attaining mAP scores of 83.08% on the DroneVehicle dataset and 77.4% on the VEDAI dataset while operating in real time. The framework demonstrates strong generalization capability across environments, establishing a new paradigm for efficient, adaptive, and robust multimodal perception in unstructured outdoor scenarios.

INDEX TERMS RGB-IR sensing, UAV remote sensing, multimodal fusion, transformer, object detection.

I. INTRODUCTION

The rapid growth of autonomous aerial systems has significantly advanced real-time traffic monitoring and public safety applications, particularly in urban and high-density environments. Relying on a single sensor modality, such as visual or infrared, can be insufficient because of complex ambient interference and variable illumination conditions [1].

Object detection based exclusively on visible or infrared imagery is limited by factors including weather conditions and inherent imaging system characteristics [2]. Visible imagery effectively captures texture and fine spatial detail under adequate illumination, whereas infrared imagery reveals thermal variations, proving advantageous in low-light scenarios [3].

The fusion of visible and infrared remote sensing images can enhance ground object recognition by leveraging their complementary information [4], [5]. Multimodal and cross-sensor methodologies have demonstrated potential across diverse sensing environments [6], [7]. This fusion approach integrates data from multiple modalities to increase accuracy in demanding operational conditions.

Recent research has explored multimodal modeling strategies for diverse visual tasks, emphasizing the promise of cross-modal representation learning in demanding contexts [8], [9]. Visible images, typically captured with low-cost, readily available CMOS sensors, are common in aerial and space platforms. The relative ease of acquiring visible remote

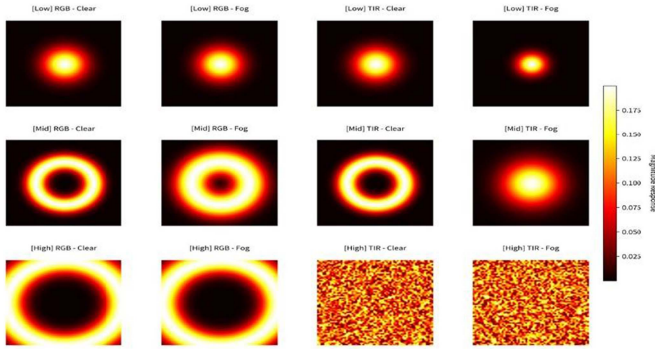


FIGURE 1. Visualization of learned frequency filters in LSD.

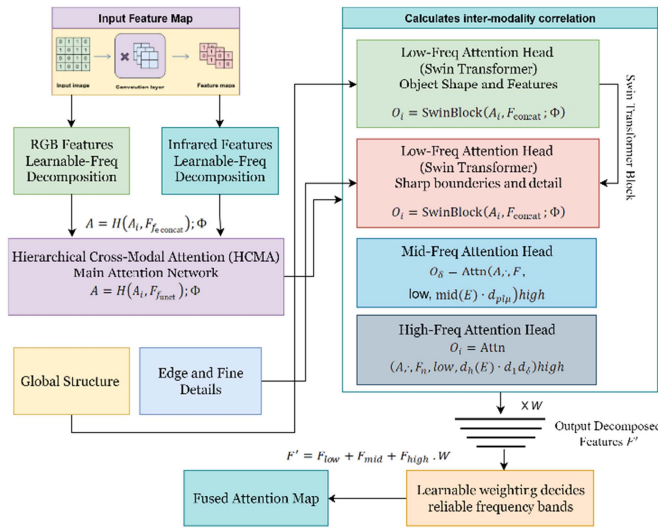


FIGURE 2. Learnable frequency decomposition (LFD).

sensing imagery has spurred rapid advances in object detection, broadening its applications. This progress is exemplified by sophisticated detection algorithms such as Faster R-CNN [10], RoITransformer [11], Gliding Vertex [12], and ReDet [13], alongside single-stage approaches including RetinaNet [14], S2ANet [15], and R3Det [16]. These developments have substantially enhanced remote sensing image interpretation.

Nevertheless, detection algorithms reliant solely on visible imagery perform inadequately under low-light or poor visibility conditions. Infrared imaging offers a distinct advantage in such scenarios, as infrared detection performance remains stable across varying illumination, being inherently unaffected by such changes. Several notable studies in infrared object detection include MSHNET [17], ACM [18], DNANet [19], and ALCNet [20].

Real-time object recognition has grown increasingly important across multiple domains, including robotics, augmented reality, video surveillance, and autonomous vehicles. Advances in detection and classification algorithms have yielded substantial improvements in accuracy [21].

The YOLO (You Only Look Once) framework has emerged as a prominent solution, noted for its effective balance of

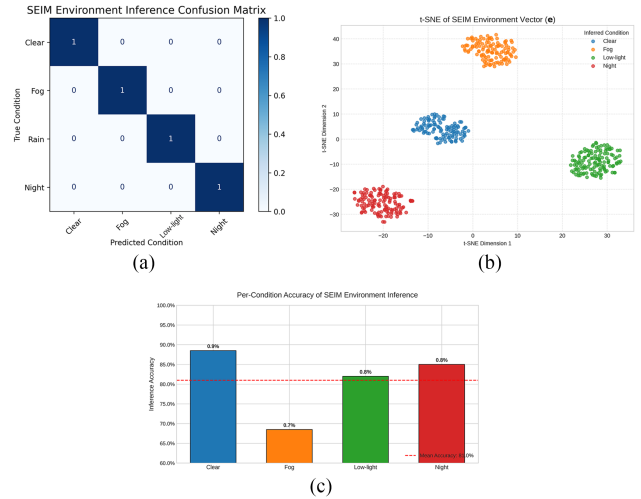


FIGURE 3. SEIM environmental inference analysis (a) representing SEIM environment inference confusion matrix, (b) representing the cluster plots for t-SNE of SEIM environment vector e and (c) pre-condition accuracy for SEIM.

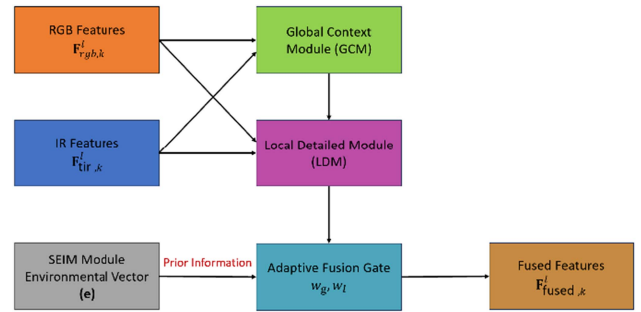


FIGURE 4. Sub-fusion expert network with prior information.

precision and processing speed, which enables efficient and accurate object detection in imagery [22], [23]. Utilizing a single-shot detection approach, YOLO divides an image into a grid and generates predictions for each grid cell. This architecture has transformed object identification by delivering high real-time performance, making it suitable for applications that demand rapid and precise detection in dynamic and complex environments [24], [25].

Despite being unaffected by illumination, infrared images demonstrate lower detection performance than visible images under well-lit conditions. Consequently, the fusion of infrared and visible imagery is gaining prominence for object detection. Current fusion-based detection systems often lack robustness in complex environments, as they typically employ uniform fusion strategies that do not account for the influence of environmental conditions on feature quality. Environmental factors can introduce significant variance in the quality of infrared and visible images, thereby affecting the complementary nature of the information they convey.

Infrared and visible images exhibit strong similarity in low-frequency features under optimal lighting and visibility. However, this similarity diminishes substantially in low-light or

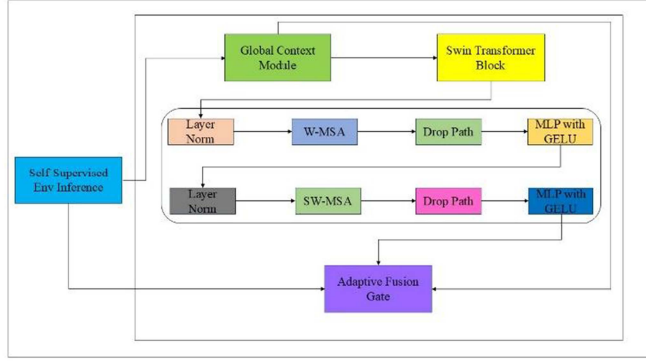


FIGURE 5. YT-SWIN core function with HCAN.

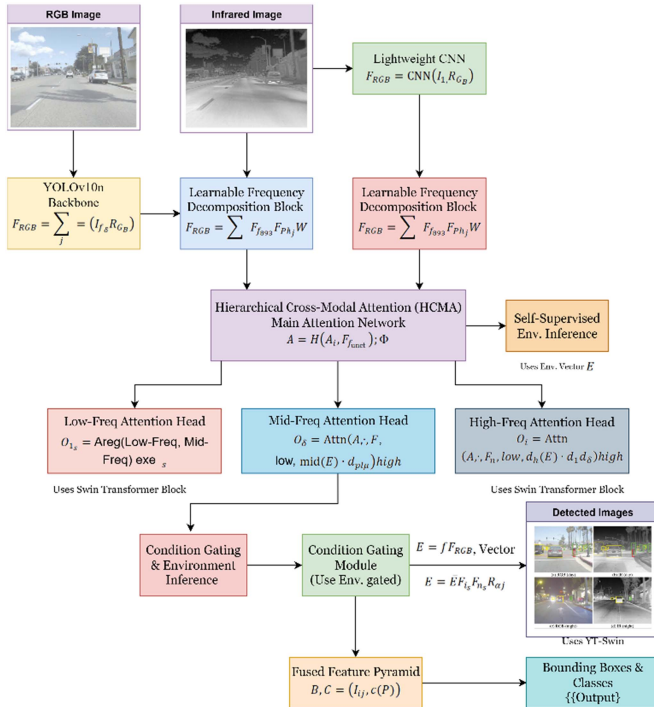


FIGURE 6. Overall YT-SWIN framework.

low-visibility scenarios. Such environmental variability creates uncertainty in the reliability and salience of multimodal features, presenting a challenge for improving detection system performance.

However, to address the computational overhead of multi-expert systems and the inherent limitations of static feature fusion in multimodal remote sensing, this paper proposes YT-SWIN, an end-to-end adaptive fusion framework. This framework dynamically recalibrates feature integration based on inferred environmental context. YT-SWIN comprises three key components.

- 1) A learnable spectral decomposition block for extracting and optimizing complementary features across frequency bands to generate more discriminative multimodal representations.

- 2) A self-supervised environment inference module that autonomously characterizes scene conditions such as weather and illumination without requiring labeled metadata.
- 3) A hierarchical cross-modal attention network, constructed upon lightweight Swin Transformer blocks. This network performs adaptive, context-aware fusion of RGB and thermal features across spatial and frequency scales, ensuring robustness across diverse environmental conditions.

By leveraging multimodal data, YT-SWIN aims to provide a more holistic understanding of traffic scenes, supporting more accurate RGB and IR image recognition and enhancing the interpretability and decision-making of intelligent UAV surveillance systems.

A. RELATED WORK

Object detection in visible and infrared remote sensing images involves fusing data from both modalities to locate ground objects. Leveraging the complementary information from these imaging sources can significantly enhance the robustness and utility of detection. Considerable research in this domain focuses on optimizing multimodal fusion at the feature level.

Sun et al. [5] introduced a visible–infrared detection framework (UA-CMDet) based on uncertainty learning. It employs cross-modal intersection-over-union (IoU) and illumination estimation to assess object uncertainty weights, reducing bias for high-uncertainty objects during training and improving fusion during inference via illumination-aware non-maximum suppression.

To address modality differences, Chen et al. [26] developed the offset-guided adaptive feature alignment (OAFA) method. It combines an offset-guided deformable alignment and fusion (ODAF) module with a cross-modality spatial offset modeling (CSOM) module to mitigate weak alignment.

Transformers have gained attention for their feature correlation modeling capabilities. Yuan and Wei [4] proposed C2Former, which uses a cross-modal attention mechanism for feature calibration to enhance intermodal complementarity. Their CAGT method employs a translation–scale–rotation alignment (TSRA) module for feature alignment and a complementary fusion transformer (CFT) module to capture complementary features, resulting in the high-performing CAGTDet detector.

He et al. [27] created CALNet, a cross-modal conflict-aware learning network designed to mitigate heterogeneity and optimize semantic fusion. Zhao et al. [28] designed the RSDet method, which uses redundant spectrum removal and dynamic feature selection modules for effective fusion and removal of interfering features. Qingyun and Zhaokui [1] developed a cross-modality attentive feature fusion (CMAFF) module, inferring separate attention maps for common and differential modalities to enhance fusion adaptability.

Sun et al. [29] proposed a specificity-guided cross-modal feature reconstruction (SCFR) approach, improving detection by focusing on distinct intermodal correlations. Sun et al.

[30] introduced the LF-MDet method, based on low-rank frequency expert encoding, to achieve high accuracy with lower computational cost.

An et al. [31] created an effectiveness-guided cross-modal information sharing network (ECISNet), incorporating a modal effectiveness guiding (MEG) module to ensure robust detection when one modality fails. Furthermore, Wang et al. [32] employed specific information improvement techniques alongside joint representation optimization to enhance fusion features' expressive quality and anti-interference capability.

The YOLO framework, a convolutional neural network (CNN)-based architecture, is widely adopted for object detection due to its effectiveness and real-time performance.

Ahmed et al. [33] performed a comparative study between YOLOv9 and YOLO-NAS for accident detection and classification. Evaluated on a dataset of 12122 annotated images using mAP@50 and mAP@50-95 metrics, the YOLOv9-c variant achieved 92.7% mAP50 and 86% mAP50-95. In contrast, the YOLO-NAS-l variant scored 85% mAP50 and 70.1% mAP50-95.

Similarly, Oleksandr Byzkrovnyi et al. [34] assessed the robustness and efficacy of YOLOv7 and Detectron2 on 300 images generated from BeamNG.drive simulations, equally split between hazardous and non-hazardous collision scenarios. Their performance evaluation reported that YOLOv7 attained 92.6% mAP@50, while Detectron2 achieved 90.9%. For mAP@50-95, Detectron2 performed better at 65.1%, compared to YOLOv7's 58.3%. Their approach utilized a deep learning model trained to classify traffic scene frames, enabling real-time multimodal object recognition.

Furthermore, the Transformer [35] is a state-of-the-art architecture and ViT [36] (Vision Transformer), the first model to perform image classification using Transformer architecture exclusively, without convolutional neural networks. When pre-trained on large-scale datasets like ImageNet-21k or JFT-300M, ViT surpassed the leading image recognition models of the time, establishing Transformer as a powerful method for computer vision.

Transformer-based multimodal object detection algorithms can be categorized by their network structure. The first category, Transformer as Backbone, employs Transformer as the core feature extractor, with a CNN often used for final prediction. Popular algorithms in this category include FPT [37], Swin Transformer [38], and DeMT [39]. The second category, Transformer as Neck, uses a CNN backbone for feature extraction and a Transformer module to refine and integrate these features before prediction. Notable algorithms in this category include DETR [40], deformable DETR [41], ACT [42], and UP-DETR [43].

Current research into visible and infrared image fusion detection employs various techniques, including uncertainty modeling, cross-modal attention mechanisms, and weak registration optimization.

However, significant challenges remain due to the substantial variation in image quality across different environments and the limited adaptability of uniform fusion strategies in

complex scenes. Furthermore, existing methods often lack a thorough analysis of frequency sub band characteristics.

To address these limitations, the YT-SWIN method is proposed in this work to meet the requirements for robust and efficient object detection across diverse scenarios. Our approach analyzes the frequency sub band features of visible and infrared remote sensing images and implements tailored fusion strategies for each, thereby effectively integrating complementary modal information and strengthening the foundation for object detection.

II. CONSTRUCTION OF YT-SWIN MODEL

A. PROBLEM STATEMENT

Accurate object detection in multimodal remote sensing requires adaptive fusion of RGB and infrared (IR) imagery, yet current techniques remain constrained. Existing methods often rely on static fusion strategies, which cannot adjust to dynamic environmental conditions such as haze, low light, or precipitation. More adaptive models, such as multi-expert or attention-based architectures, introduce high computational overhead and frequently depend on manually annotated environmental metadata, limiting their practical applicability. Consequently, a lightweight, end-to-end fusion framework capable of autonomously inferring scene conditions and dynamically recalibrating cross-modal integration is clearly needed to maintain robustness without sacrificing efficiency.

To address these issues, we propose improvement strategies from the following aspects.

- 1) We introduce a self-supervised environment inference module that automatically characterizes scene conditions directly from input data, eliminating the requirement for labeled environmental metadata.
- 2) We propose a hierarchical cross-modal attention network constructed from Swin Transformer blocks to address static, non-adaptive fusion. This architecture enables dynamic, context-aware recalibration of RGB-TIR fusion by effectively modeling both local and global feature relationships.
- 3) We leverage the linear computational complexity of the Swin Transformer and integrate an adaptive channel pruning strategy within a multi-scale YOLOv10n backbone. This design maintains real-time efficiency while ensuring robustness, achieving a balance between performance and computational economy.

III. MODELING STRATEGIES

A. DUAL STREAM FEATURE EXTRACTION

The primary RGB processing stream utilizes the YOLOv10n, specifically utilized for nano-scale edge deployment. YOLOv10n incorporates as backbone with structural reparameterization for optimal speed-accuracy trade-off. Given an input RGB image $\mathbf{I}_{rgb} \in \mathbb{R}^{H \times W \times 3}$, the backbone $f_{rgb}(\cdot)_{\text{extracts}}$ multi-scale features through three hierarchical stages:

$$\mathbf{I}_{rgb}^l = f_{rgb}^l(\mathbf{I}_{rgb}), l \in \{1, 2, 3\} \quad (1)$$

where $\mathbf{F}_{rgb}^1 \in \mathbb{R}^{C_1 \times H_1 \times W_1}$, $\mathbf{F}_{rgb}^2 \in \mathbb{R}^{C_2 \times H_2 \times W_2}$, and $\mathbf{F}_{rgb}^3 \in \mathbb{R}^{C_3 \times H_3 \times W_3}$ represent features at progressively coarser spatial resolutions with increased channel dimensions. Each stage employs RepConv blocks during training:

$$\mathbf{F}_{out}^l = \text{BN}(\mathbf{W}_1^l * \mathbf{F}_{in}^l) + \text{BN}(\mathbf{W}_2^l * \mathbf{F}_{in}^l) + \text{Identity}(\mathbf{F}_{in}^l) \quad (2)$$

where $*$ denotes convolution, BN represents Batch Normalization, and $\mathbf{W}_1^l, \mathbf{W}_2^l$ are learnable convolutional weights. During inference, these parallel branches merge into single convolutional layers through structural reparameterization:

$$\mathbf{W}_{merged}^l = \mathbf{W}_1^l + \mathbf{W}_2^l, \mathbf{b}_{merged}^l = \mathbf{b}_1^l + \mathbf{b}_2^l \quad (3)$$

This reparameterization reduces computational complexity by approximately 40% while maintaining representational capacity. The backbone further incorporates Efficient Channel Attention (ECA) modules after each RepConv block:

$$\mathbf{F}_{eca} = \sigma(\text{Conv}_{1D}(\text{GAP}(\mathbf{F}))) \otimes \mathbf{F} \quad (4)$$

where GAP denotes Global Average Pooling, σ is the sigmoid function, and $\otimes$ represents channel-wise multiplication.

For thermal imagery $\mathbf{I}_{tir} \in \mathbb{R}^{H \times W}$, we employ a lightweight ShuffleNetV2 2×0.5 encoder $f_{tir}(\cdot)$ designed specifically for efficiency:

$$\mathbf{F}_{tir}^l = f_{tir}^l(\mathbf{I}_{tir}), l \in \{1, 2, 3\} \quad (5)$$

The encoder utilizes channel shuffle operations to enable information flow across channels without dense convolutions:

$$\mathbf{F}_{shuffled} = \text{Shuffle}(\text{Concat}(\text{DWConv}(\mathbf{F}_1), \text{DWConv}(\mathbf{F}_2))) \quad (6)$$

where DWConv denotes depth wise separable convolution, and $\mathbf{F}_1, \mathbf{F}_2$ represent split feature groups. This design maintains a computational budget under 100 MFLOPs while preserving multi-scale feature extraction capability.

For feature pyramid construction the following feature extraction, both streams construct feature pyramids using a modified PANet-Lite architecture that replaces standard convolutions with Ghost Modules:

$$\mathbf{P}_l^{up} = \text{NearestUpsample}(\mathbf{P}_{l+1}) \oplus \text{GhostConv}_{1 \times 1}(\mathbf{F}_l) \quad (7)$$

$$\mathbf{P}_l^{down} = \text{MaxPool}_{2 \times 2}(\mathbf{P}_{l-1}) \oplus \text{GhostConv}_{3 \times 3}(\mathbf{F}_l) \quad (8)$$

where GhostConv generates feature maps through cheap linear operations:

$$\mathbf{Y} = \Phi(\mathbf{X} * \mathbf{W}_1) \oplus (\mathbf{X} * \mathbf{W}_2) \quad (9)$$

with $\Phi_{representing}$ depth wise separable convolution, reducing parameters by approximately 50% compared to standard convolutions.

B. LEARNABLE SPECTRAL DECOMPOSITION BLOCK

Building upon frequency-domain analysis principles, we propose a Learnable Spectral Decomposition Block (LSD) that

adaptively separates features into task-relevant frequency bands. Given intermediate features $\mathbf{F}_m^l \in \mathbb{R}^{C \times H \times W}$ from modality $m \in \{rgb, tir\}$, we first compute the 2D Discrete Fourier Transform (DFT) independently per channel:

$$\mathcal{F}_m^l(u, v) = \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} \mathbf{F}_m^l(h, w) \cdot e^{-2\pi j(\frac{uh}{H} + \frac{vw}{W})} \quad (10)$$

The resulting complex-valued representation $\mathcal{F}_m^l = \mathcal{F}_m^{l, \text{real}} + j\mathcal{F}_m^{l, \text{imag}}$ is decomposed into amplitude $\mathbf{A}_m^l$ and phase Φ_m^l components:

$$\mathbf{A}_m^l = \sqrt{(\mathcal{F}_m^{l, \text{real}})^2 + (\mathcal{F}_m^{l, \text{imag}})^2} \quad (11)$$

$$\Phi_m^l = \arctan\left(\frac{\mathcal{F}_m^{l, \text{imag}}}{\mathcal{F}_m^{l, \text{real}}}\right) \quad (12)$$

learnable complex-valued filters $\mathbf{W}_k \in \mathbb{C}^{C \times C \times K}$ parameterized in the frequency domain:

$$\mathcal{F}_{m,k}^l(u, v) = \mathcal{F}_m^l(u, v) \odot \mathbf{W}_k(u, v) + \mathbf{B}_k(u, v) \quad (13)$$

where $\odot$ denotes Hadamard product, $\mathbf{B}_k \in \mathbb{C}^C$ are learnable complex biases, and $K = 3$ represents our decomposition into low, medium, and high-frequency bands. The inverse DFT reconstructs spatially filtered features:

$$\mathbf{F}_{m,k}^l(h, w) = \frac{1}{HW} \sum_{u=0}^{H-1} \sum_{v=0}^{W-1} \mathcal{F}_{m,k}^l(u, v) \cdot e^{2\pi j(\frac{uh}{H} + \frac{vw}{W})} \quad (14)$$

To enforce stability and meaningful decomposition, we apply spectral sparsity regularization:

$$\mathcal{L}_{\text{spectral}} = \lambda_s \sum_{k=1}^K \|\mathbf{w}_k\|_1 + \lambda_o \sum_{k=1}^{K-1} \sum_{k'=k+1}^K \|\mathbf{w}_k \odot \mathbf{w}_{k'}\|_F \quad (15)$$

This yields modality-specific frequency-decomposed feature sets $\{\mathbf{F}_{rgb,k}^l\}_{k=1}^K$ and $\{\mathbf{F}_{tir,k}^l\}_{k=1}^K$ for subsequent fusion.

The amplitude components primarily capture structural information (low-frequency content), while phase components preserve spatial relationships and high-frequency details. Fig. 1 visualizes the magnitude response of the learning optimal filters $\mathbf{W}_k$ for low, mid and high frequency bands, the network adaptively learns distinct spectral profiles, for RGB in clear conditions, high frequency components are amplified (highlighting edges), while for IR in fog, low frequency components are emphasized (preserving thermal blobs). The magnitude response in heatmap where the center represents low frequencies and the periphery represents high frequencies. This validates that the LSD block learns physically plausible decompositions tailored to each sensor and environment.

The complete spectral decomposition process for each modality can be expressed as:

$$\{\mathbf{F}_{m,k}^l\}_{k=1}^K = \text{LSD}(\mathbf{F}_m^l; \{\mathbf{W}_k\}) \quad (16)$$

where LSD denotes our Learnable Spectral Decomposition Block shown in Fig. 2. This data-adaptive approach enables more effective feature separation than fixed transformations, providing optimized representations for subsequent cross-modal fusion.

C. SELF-SUPERVISED ENVIRONMENTAL INFERENCE MODULE

To enable environment-adaptive fusion without requiring manually labeled metadata, we propose a Self-Supervised Environment Inference Module (SEIM) that learns to characterize scene conditions directly from input feature statistics. Drawing inspiration from contrastive representation learning, our module operates on the hypothesis that distinct environmental conditions produce characteristic distributions in low-level multimodal feature representations.

Given early-layer features from both modalities, $\mathbf{F}_{rgb}^1 \in \mathbb{R}^{C_1 \times H_1 \times W_1}$ and $\mathbf{F}_{tir}^1 \in \mathbb{R}^{C_1 \times H_1 \times W_1}$, we first compute channel-wise statistics that serve as environment-sensitive descriptors. For each feature map, we extract mean μ_m and standard deviation σ_m vectors:

$$\mu_m = \frac{1}{H_1 W_1} \sum_{h=1}^{H_1} \sum_{w=1}^{W_1} \mathbf{F}_m^1(:, h, w) \quad (17)$$

$$\sigma_m = \sqrt{\frac{1}{H_1 W_1} \sum_{h,w} (\mathbf{F}_m^1(:, h, w) - \mu_m)^2} \quad (18)$$

These statistics are concatenated to form a joint environment descriptor vector $\mathbf{d} \in \mathbb{R}^{4C_1}$:

$$\mathbf{d} = [\mu_{rgb}; \sigma_{rgb}; \mu_{tir}; \sigma_{tir}] \quad (19)$$

A lightweight inference network $g_\theta(\cdot)$, composed of two fully-connected layers with ReLU activation, processes this descriptor to produce an environment context vector $\mathbf{e} \in \mathbb{R}^K$:

$$\mathbf{e} = \text{Softmax}(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \mathbf{d} + \mathbf{b}_1) + \mathbf{b}_2) \quad (20)$$

where $\mathbf{W}_1 \in \mathbb{R}^{D \times 4C_1}$, $\mathbf{W}_2 \in \mathbb{R}^{K \times D}$ are learnable weight matrices, and K represents the number of latent environmental clusters (e.g., clear, fog, rain, night, snow). To train this module without explicit labels, we employ a self-supervised contrastive objective that leverages natural environmental variations within training batches. For a batch of N_{image} pairs, we first compute their environment vectors $\{\mathbf{e}_i\}_{i=1}^N$. During training, we maintain a dynamic queue $\mathcal{Q}$ of previously computed environment vectors as negative samples. The contrastive loss maximizes agreement between differently augmented views of the same input while minimizing similarity to other samples:

$$\mathcal{L}_{\text{env}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\mathbf{e}_i \cdot \mathbf{e}_i^+ / \tau)}{\exp(\mathbf{e}_i \cdot \mathbf{e}_i^+ / \tau) + \sum_{\mathbf{e}_j^- \in \mathcal{Q}} \exp(\mathbf{e}_i \cdot \mathbf{e}_j^- / \tau)} \quad (21)$$

where $\mathbf{e}_i^+$ is a positive pair generated through weak imagery augmentation of the input pair, $\mathbf{e}_j^-$ denotes negative samples from the queue, and τ is a temperature hyperparameter controlling distribution sharpness. The queue $\mathcal{Q}$ is updated using a first-in-first-out strategy with momentum encoding to ensure stable training. Additionally, we introduce a consistency regularization term that encourages the environment inference to be scale-invariant across different network depths:

$$\mathcal{L}_{\text{consist}} = \frac{1}{L-1} \sum_{l=2}^L \|\mathbf{e}^l - \mathbf{e}^{l-1}\|_2^2 \quad (22)$$

where $\mathbf{e}^l$ represents environment vectors computed from features at layer l . This regularization ensures that environmental characterization remains coherent throughout the network hierarchy. Fig. 3 illustrating the inferred environment vector $\mathbf{e}$ serves as a conditioning signal for downstream fusion modules, enabling them to dynamically adjust their behavior according to the detected environmental conditions. a mean accuracy of 81% (with minor variations across classes), which underscores the robustness and stability of the inference process. Collectively, these results confirm that SEIM effectively captures and separates the underlying physical properties of diverse environmental conditions, providing a reliable foundation for downstream adaptation. This approach eliminates dependency on manually labeled environmental metadata while providing adaptive capabilities crucial for all-weather perception systems.

D. HIERARCHICAL CROSS-MODEL ATTENTION NETWORK

The Hierarchical Cross-Modal Attention Network (HCAN) constitutes the central fusion mechanism, engineered to integrate both global contextual semantics and local spatial detail from RGB and thermal modalities under a unified, environment-adaptive framework. The network first transforms the frequency-decomposed feature pairs $\{\mathbf{F}_{rgb,k}^l, \mathbf{F}_{tir,k}^l\}$ for each scale l and frequency band k via a Conv Module, which applies a stride-2 convolution $\text{Conv}_{1 \times 1}^{(s=2)}$ followed by Batch Normalization (BN) and a Silu activation function:

$$\mathbf{X}_{m,k}^l = \text{Silu} \left(\text{BN} \left(\text{Conv}_{1 \times 1}^{(s=2)} \left(\mathbf{F}_{m,k}^l \right) \right) \right), \quad m \in \{rgb, tir\} \quad (23)$$

This operation reduces spatial dimensions shown in Table 1 while expanding channel capacity for subsequent processing. The transformed feature $\mathbf{X}_{m,k}^l$ is then decomposed through three parallel 1×1 convolutional expansions into distinct representation streams:

$$\mathbf{H}_{m,k}^{l,(1)}, \mathbf{H}_{m,k}^{l,(2)}, \mathbf{H}_{m,k}^{l,(3)} = \text{Split} \left(\text{Conv}_{1 \times 1}^{(3)} \left(\mathbf{X}_{m,k}^l \right) \right) \quad (24)$$

Here, $\mathbf{H}_{m,k}^{l,(1)}$ serves as a residual connection, $\mathbf{H}_{m,k}^{l,(2)}$ is routed to the Global Context Module (GCM), and $\mathbf{H}_{m,k}^{l,(3)}$ is directed to the Local Detail Module (LDM) shown in Fig. 4. The GCM captures long-range dependencies essential for understanding

TABLE 1. Parameters Description

Parameters	Symbol	Value
Input Channel	C_{in}	254
Number of Heads	h	8
Window Size	M	7
MLP Expansion Ratio	r	4
Number of Blocks	n	2
Hidden Dimensions	C_{hidden}	1024
Head Dimension	d_{head}	32
Bias Table Size	B_{size}	$(2M-1) \times (2M-1)$
Total Parameters	N_{params}	$\sim 12.8M$
Complexity	O	$O(M^2 \times HW \times C)$

complex remote sensing backgrounds through a dual-path self-attention mechanism. The channel attention path is computed as:

$$\mathbf{A}_{ch} = \sigma \left(\text{MLP} \left(\text{GAP} \left(\mathbf{H}_{m,k}^{l,(2)} \right) \right) + \text{MLP} \left(\text{GMP} \left(\mathbf{H}_{m,k}^{l,(2)} \right) \right) \right) \otimes \mathbf{H}_{m,k}^{l,(2)} \quad (25)$$

while the spatial attention path aggregates multi-scale context via:

$$\mathbf{A}_{sp} = \sigma \left(\text{Conv}_{7 \times 7} \left(\left[\text{AvgPool} \left(\mathbf{H}_{m,k}^{l,(2)} \right); \text{MaxPool} \left(\mathbf{H}_{m,k}^{l,(2)} \right) \right] \right) \right) \otimes \mathbf{H}_{m,k}^{l,(2)} \quad (26)$$

where GAP and GMP denote Global Average and Max Pooling, σ is the sigmoid function, and $\otimes$ represents element-wise multiplication. The globally-enhanced feature is then obtained as:

$$\mathbf{G}_{m,k}^l = \text{Conv}_{3 \times 3} \left(\left[\mathbf{A}_{ch}; \mathbf{A}_{sp} \right] \oplus \mathbf{H}_{m,k}^{l,(2)} \right) \quad (27)$$

Simultaneously, the LDM, constructed from n stacked Swin Transformer blocks, models fine-grained local interactions through windowbased multi-head self-attention. For multimodal inputs, cross-attention between RGB and thermal features is computed as:

$$\mathbf{L}_k^l = \text{SwinBlock}_n \left(\left[\mathbf{H}_{rgb,k}^{l,(3)}; \mathbf{H}_{tir,k}^{l,(3)} \right] \right) \quad (28)$$

with each SwinBlock implementing the following operations for layer l :

$$\hat{\mathbf{z}}^l = \mathbf{W} - \text{MSA} \left(\text{LN} \left(\mathbf{z}^{l-1} \right) \right) + \mathbf{z}^{l-1} \quad (29)$$

$$\mathbf{z}^l = \text{MLP} \left(\text{LN} \left(\hat{\mathbf{z}}^l \right) \right) + \hat{\mathbf{z}}^l \quad (30)$$

$$\hat{\mathbf{z}}^{l+1} = \text{SW} - \text{MSA} \left(\text{LN} \left(\mathbf{z}^l \right) \right) + \mathbf{z}^l \quad (31)$$

$$\mathbf{z}^{l+1} = \text{MLP} \left(\text{LN} \left(\hat{\mathbf{z}}^{l+1} \right) \right) + \hat{\mathbf{z}}^{l+1} \quad (32)$$

An environment-aware gating mechanism, conditioned on the inferred environment vector $\mathbf{e}$, dynamically weights the contributions of the global and local pathways.

The fusion weights are computed via:

$$\mathbf{w}_g^l, \mathbf{w}_l^l = \text{Softmax} \left(\text{MLP} \left(\mathbf{e} \oplus \text{GAP} \left(\mathbf{X}_{m,k}^l \right) \right) \right) \quad (33)$$

The weighted outputs from the GCM and LDM are then combined with the residual stream and fused through a final convolution:

$$\mathbf{F}_{fused,k}^l = \text{Conv}_{3 \times 3} \left(\mathbf{H}_{m,k}^{l,(1)} \oplus \left(\mathbf{w}_g^l \otimes \mathbf{G}_{m,k}^l \right) \oplus \left(\mathbf{w}_l^l \otimes \mathbf{L}_{m,k}^l \right) \right) \quad (34)$$

This adaptive fusion is performed across all frequency bands and spatial scales, with multi-scale aggregation further refined through learnable up sampling and down sampling coefficients. The final integrated feature at scale l_{is} obtained via:

$$\mathbf{F}_{final}^l = \sum_{k=1}^K \gamma_k^l \cdot \text{Upsample} \left(\mathbf{F}_{fused,k}^l \right) + \delta_k^l \cdot \text{Downsample} \left(\mathbf{F}_{fused,k}^{l+1} \right) \quad (35)$$

where γ_k^l and δ_k^l are parameters learned during training. By unifying global scene understanding with local feature precision in a computationally efficient, parallelized architecture, HCAN in Fig. 5 significantly enhances the robustness and accuracy of object detection in diverse and cluttered remote sensing environments.

E. DETECTION HEAD AND OPTIMIZATION

The fused feature pyramid $\{\mathbf{F}_{fused}^l\}_{l=1}^3$ feeds into YOLOv10n's detection head, which employs anchor-free detection with decoupled head for classification and regression:

$$\mathbf{P}_{cls}^l, \mathbf{P}_{reg}^l = \text{DecoupledHead} \left(\bar{\mathbf{F}}_{fused}^l \right) \quad (36)$$

The training objective combines detection, environment, and spectral losses:

$$\mathcal{L}_{total} = \lambda_{det} \mathcal{L}_{det} + \lambda_{env} \mathcal{L}_{env} + \lambda_{spec} \mathcal{L}_{spectral} \quad (37)$$

With $\mathcal{L}_{det} = \mathcal{L}_{DFL} + \mathcal{L}_{CIoU} + \mathcal{L}_{cls}$, where DFL (Distribution Focal Loss) improves localization accuracy, CIoU (Complete IoU) enhances bounding box regression, and classification loss employs Quality Focal Loss for handling class imbalance. This integrated methodology enables YT-SWIN to achieve state-of-the-art performance in multimodal object detection while maintaining real-time efficiency through its optimized YOLOv10n-based architecture and adaptive fusion mechanisms.

IV. YT-SWIN MODEL DESIGN

A. OVERALL SYSTEM DESIGN

The YT-SWIN farmwork adopts a dual stream encoder-decoder architecture optimized for real-time multimodal object detection under diverse environmental conditions. As

illustrated in Fig. 6, the system process RGB and Thermal Infrared (TIR) inputs through parallel feature extraction pathways, followed by adaptive frequency-aware fusion continued on self-supervised environmental inference, culminating in a unified YOLOv10n-based detection head. The architecture comprises four principal components: (1) A YOLOv10n-based RGB encoder with complementary lightweight TIR encoder, (2) A learnable Spectral decomposition block (LSD) for frequency-domain feature separation, (3) A self-supervised Environmental Inference Module (SEIM), and (4) A Hierarchical Cross-Modal Attention Network (HCAN) build with Swin Transformer blocks for dynamic feature fusion. The complete pipeline is designed for end-to-end training and optimized for deployment in resource-constrained UAV remote sensing platforms.

B. ALGORITHM'S PSEUDOCODE

YT-SWIN Pipeline

Input: RGB image $I_{rgb} \in \mathbb{R}^{H \times W \times 3}$, Thermal image $I_{tir} \in \mathbb{R}^{H \times W}$
Output: Bounding boxes B , Class labels C , Confidence scores S

- 1: Dual Stream Feature Extraction
- 2: $F_{rgb} \leftarrow f_{YOLOv10n}(I_{rgb})$
- 3: $F_{tir} \leftarrow f_{LightCNN}(I_{tir})$
- 4: 2. Learnable Spectral Decomposition
- 5: for each modality $m \in \{rgb, tir\}$ do
- 6: $F_{freq} \leftarrow FFT2D(F_m)$
- 7: for $k \in \{low, mid, high\}$ do
- 8: $F_m^k \leftarrow IFFT2D(F_m \odot W_k)$
- 9: end for
- 10: end for
- 11: 3. Self-Supervised Environment Inference
- 12: $d \leftarrow [\mu(F_{rgb}), \sigma(F_{rgb}), \mu(F_{tir}), \sigma(F_{tir})]$
- 13: $e \leftarrow Softmax(g_\phi(d))$
- 14: 4. Hierarchical Cross-Modal Attention Fusion
- 15: for each scale $l \in \{1, 2, 3\}$ do
- 16: for each frequency band k do
- 17: //Global Context Module
- 18: $A_{ch} \leftarrow ChannelAttention(F_{rgb,k}^l, F_{tir,k}^l)$
- 19: $A_{sp} \leftarrow SpatialAttention(F_{rgb,k}^l, F_{tir,k}^l)$
- 20: $G \leftarrow Conv_{3 \times 3}([A_{ch}; A_{sp}])$
- 21: //Swin Transformer Module
- 22: $z_0 \leftarrow LayerNorm([F_{rgb,k}^l; F_{tir,k}^l])$
- 23: for n blocks do
- 24: //W-MSA (even blocks)
- 25: $z' \leftarrow WindowMSA(z, window_size=7, heads=8)$
- 26: $z \leftarrow DropPath(z') + z$
- 27: $z \leftarrow MLP(LayerNorm(z)) + z$
- 28: //SW-MSA (odd blocks)
- 29: $z'' \leftarrow ShiftedWindowMSA(z, shift=(3,3), heads=8)$
- 30: $z \leftarrow DropPath(z'') + z$

- 31: $z \leftarrow MLP(LayerNorm(z)) + z$
- 32: end for
- 43: $L \leftarrow z$
- 34: //Environment-Aware Fusion
- 35: $w_g, w_l \leftarrow Softmax(h_\psi([e; GAP(G); GAP(L)]))$
- 36: $F_{fused}^k \leftarrow Conv_{3 \times 3}(F_{rgb,k}^l + w_g \otimes G + w_l \otimes L)$
- 37: end for
- 38: //Multi-scale aggregation
- 39: $P^l = \sum_{k=1}^3 (\gamma_k^l \cdot Upsample(F_{fused}^k) + \delta_k^l \cdot Downsample(F_{fused}^{k+1}))$
- 40: end for
- 41: 5. Detection Head
- 42: $B, C, S \leftarrow YOLOv10n_{DetectionHead}(F_{final})$
- 43: return B, C, S

V. EXPERIMENTAL ANALYSIS

To validate the proposed UAV traffic monitoring framework, we conduct experimental analysis using traffic datasets from Kaggle. YT-SWIN model's evaluation is conducted to validate its efficacy in multimodal remote sensing object detection under diverse environmental conditions. We perform comprehensive experiments on two benchmark datasets, DroneVehicle and VEDAI under clear, foggy rainy, and low light scenarios. The proposed method is compared against state-of-the-art approaches, with ablation studies analyzing the contribution of each component. Implementation details, evaluation metrics, and quantitative and qualitative results are systematically presented in the following subsections.

A. DATASETS AND PROTOCOL DESCRIPTION

We evaluate our model on the two different datasets, the DroneVehicle [44] and VEDAI [45] datasets for training and validating the proposed method.

The DroneVehicle dataset is a comprehensive UAV-based remote sensing dataset specifically designed for vehicle detection tasks, comprising paired visible and infrared images. It contains 28439 image pairs with oriented bounding boxes, totaling 953087 annotated instances across five categories: car, bus, truck, van, and freight car. The dataset covers diverse scenarios including urban roads, residential areas, and highways, with imagery captured from various altitudes and viewing angles during both daytime and nighttime. To ensure robust evaluation with balanced category distributions and uniform sampling across viewpoints, the dataset is partitioned into training (17990 pairs), validation (1469 pairs), and testing (8980 pairs) subsets.

The VEDAI dataset consists of 1246 small-sized images from diverse locations, including grasslands, highways, mountainous regions, and urban areas. Each image is provided at two resolutions: 1024×1024 and 512×512 pixels. Every scene contains paired RGB and infrared (IR) modalities, with all images captured during daytime at a 90° vertical viewing angle. The dataset comprises 3640 annotated vehicle

instances across 11 categories, such as cars, boats, campers, and airplanes. For evaluation, the dataset is divided into ten cross-validation folds, each containing 1089 training images and 121 test images.

B. EVALUATION METRICS SETUP

The primary evaluation metrics employed are two mean average precision (mAP) measures mAP50. Additionally, model complexity and computational efficiency are assessed using the number of parameters (Params) and giga floating-point operations (GFLOPs), respectively.

The calculation of mAP is based on the Precision-Recall (PR) curve, which comprehensively illustrates a model's detection performance across all categories. A higher mAP value indicates superior overall detection performance.

Precision (P) measures the accuracy of positive predictions, while Recall (R) quantifies the proportion of actual positives correctly identified. The intersection-over-union (IoU) threshold defines the basis for these metrics. Specifically, mAP50 represents the mean average precision across all categories at an IoU threshold ≥ 0.5 . In contrast, mAP50:95 is the mean average precision calculated over a range of IoU thresholds from 0.5 to 0.95 with a step size of 0.05.

Intersection-over-union (IoU) is computed between a predicted bounding box (Bp) and its corresponding ground-truth bounding box (Bgt). Predictions are categorized as follows: True Positive (TP) indicates a correctly identified positive sample; False Positive (FP) indicates a negative sample mistakenly identified as positive; False Negative (FN) indicates a positive sample the model failed to detect; and True Negative (TN) indicates a correctly identified negative sample.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (38)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (39)$$

$$\text{IoU} = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|} \quad (40)$$

$$AP_i = \int_0^1 P_i(r) dr \quad (41)$$

The Average Precision (AP_i) for a class i is calculated by integrating its PR curve. The mean Average Precision (mAP) is then the arithmetic mean of the AP values across all n classes.

C. IMPLEMENTATION SETUP DETAIL

All experiments in this work were conducted on an NVIDIA GeForce RTX A6000 GPU with a python environment, using CUDA 11.7 and PyTorch. For the DroneVehicle dataset, we employed a baseline method (YOLOv10n) that utilizes a Siamese backbone and an additive fusion strategy. The model was trained for 100 epochs using the SGD optimizer, with input images sized at 640×640 pixels, an initial learning rate of 0.01, and a weight decay of 0.01 shown in Table 2. For the VEDAI dataset, given its small object sizes, the baseline

TABLE 2. Training Parameters for YT-SWIN Model

Hyperparameters	Value
Batch size	16
Learning rates	0.0002
Iou	0.7
momentum	0.937
Weight-decay	0.01
Optimizer	Adam
Label	0.1
Epochs	100
Loss	entropy

used was YOLOv10n, also based on a Siamese backbone and additive fusion.

This model was trained for 30 epochs using the same optimizer parameters (initial learning rate of 0.0002 and weight decay of 0.01) and the same input image resolution of 640×640 pixels.

Evaluation on the DroneVehicle dataset followed the protocol established, where performance was measured using mean average precision (mAP) at an Intersection-over-Union (IoU) threshold of 0.5. Detection accuracy for the VEDAI dataset was evaluated using the COCO-compliant assessment pipeline which calculates the average precision (AP) over a range of IoU thresholds from 0.5 to 0.95 at intervals of 0.05, as required.

D. EVALUATION WITH STATE-OF-THE-ART METHODS IN DRONEVEHICLE

To comprehensively evaluate the performance of the proposed YT-SWIN method Fig. 8, we conducted extensive comparisons with several state-of-the-art object detection algorithms. The assessment includes ten advanced visible-infrared detection techniques including prominent models such as CALNet and C2Former alongside five single-modality detection algorithms as benchmark references. The detailed comparative results are summarized in Fig. 7. Experimental results demonstrate that the proposed YT-SWIN approach achieves a mAP score of 83.08%, outperforming all other compared methods. Its frequency-domain fusion and detection framework for visible-infrared data yields a 3.68% improvement in mAP over the second-best method (DMM), indicating robust performance in complex scenarios, and a 7.69% gain over CALNet. YT-SWIN also exhibits enhanced fusion feature representation, with mAP improvements of 6.15% and 8.85% over FFODNet and C2Former, respectively. For key object categories such as "car," "bus," and "truck," the proposed method consistently produces higher mAP scores than existing approaches. These findings validate the effectiveness of the YT-SWIN in enhancing the capability of remote sensing object detection algorithms to handle challenging environments.

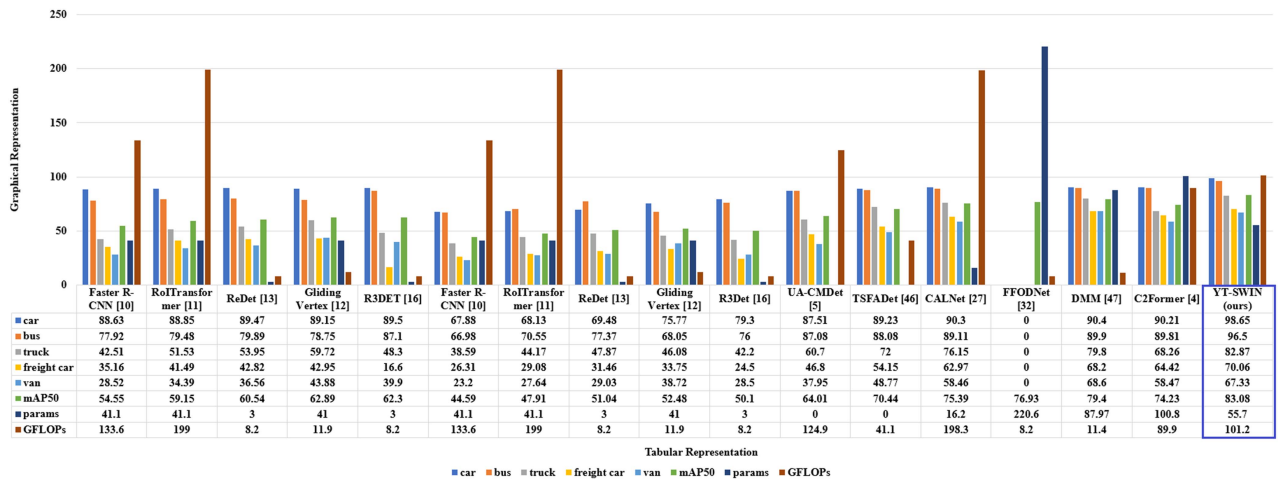


FIGURE 7. Evaluation results of different detection methods in DroneVehicle. The best performance measurement of our model is highlighted in Blue.

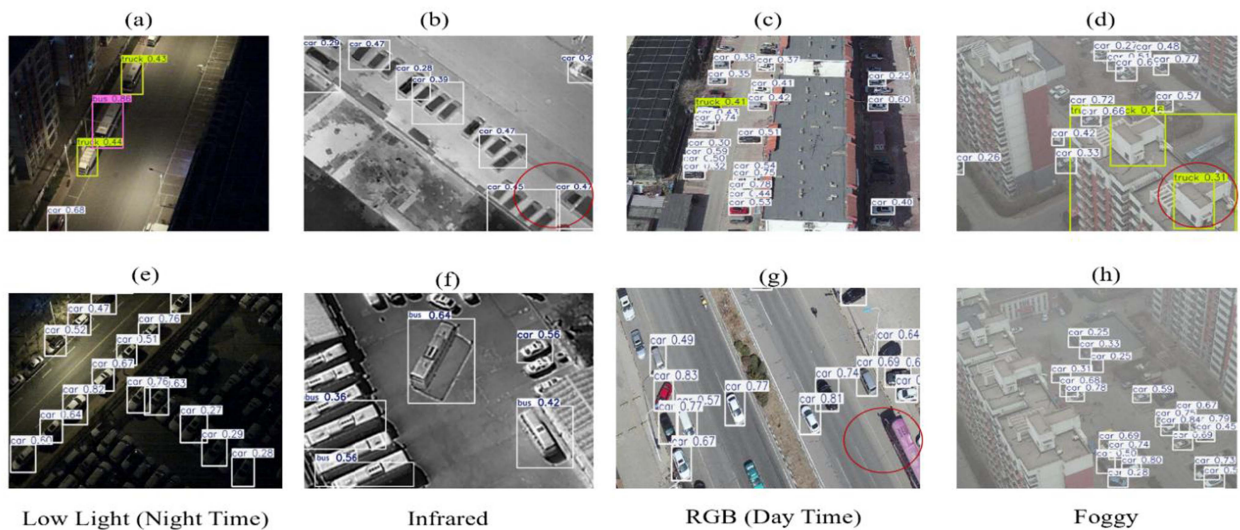


FIGURE 8. Results of object detection in dronevehicle dataset by YT-SWIN fusion model in IR, RGB, and foggy environment and the red circles represent the failure analysis in detection results.

E. EVALUATION WITH STATE-OF-THE-ART METHODS IN VEDAI

The quantitative analysis results on the VEDAI dataset are evaluated in Fig. 9. A ten-fold cross-validation was conducted, comparing the proposed multimodal model YT-SWIN against single-modal baselines YOLOv4, YOLOv8n, and other multimodal models including YOLOv8n, YOLOv10n, YOLOv8s, YOLOrs, Super-YOLO, and MF-YOLO. YT-SWIN achieved a final mAP50 of 77.4%. Compared to the RGB and IR single-modal versions, YT-SWIN improved mAP50 by 10%, respectively. Furthermore, it outperformed the multimodal YOLOv10n baseline model by a 3.7% gain in mAP50. Against other advanced models, YT-SWIN surpassed Super-YOLO in most categories, with notable improvements of 11% for *Boat* and 3.9% for *Truck*. While overall mAP50 was comparable to MF-YOLO, it was slightly less accurate on small objects such as *Car*, *Truck*, *Tractor*, and *Van*. However, it demonstrated superior performance for *Truck*, *Car*, and

Camper. While YT-SWIN achieves a slightly higher mAP across all classes small to large objects compare to MF-YOLO's on VEDAI.

Visual results on the VEDAI dataset are illustrated in Fig. 10. YT-SWIN outperformed the baseline model in complex scenes, showing greater accuracy in distinguishing foreground from background in urban, grassland, and road traffic environments. It exhibited particularly high detection precision for challenging categories like *car*, which is frequently confused with the background. In simpler road scenes, where capturing tiny object details is critical, YT-SWIN effectively utilized multimodal information to reduce the false detection rate.

The proposed framework demonstrates universal robustness across diverse object characteristics and environmental complexities; we strategically evaluated its performance on two complementary benchmark datasets. The DroneVehicle dataset was employed to validate the model's efficacy in

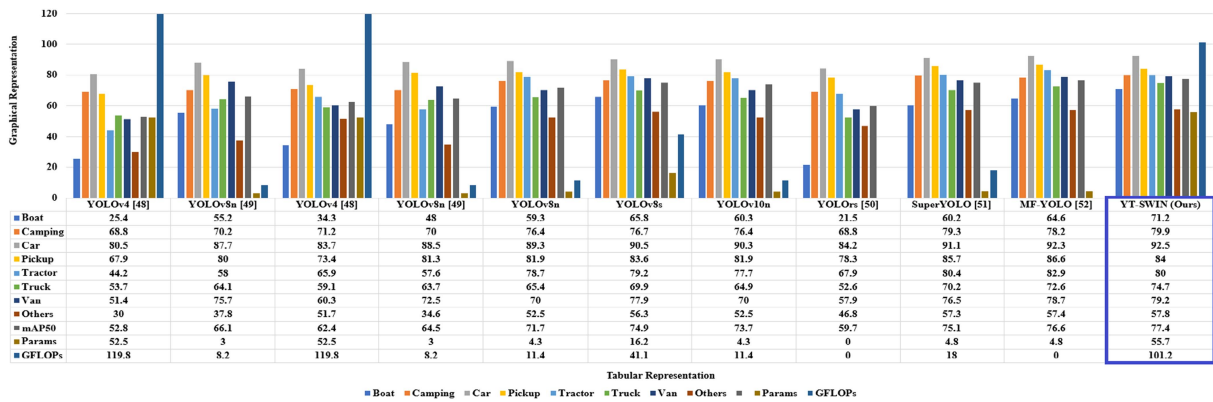


FIGURE 9. Evaluation results of different detection methods in VEDAI. The best performance measurement of our model is highlighted in Blue.

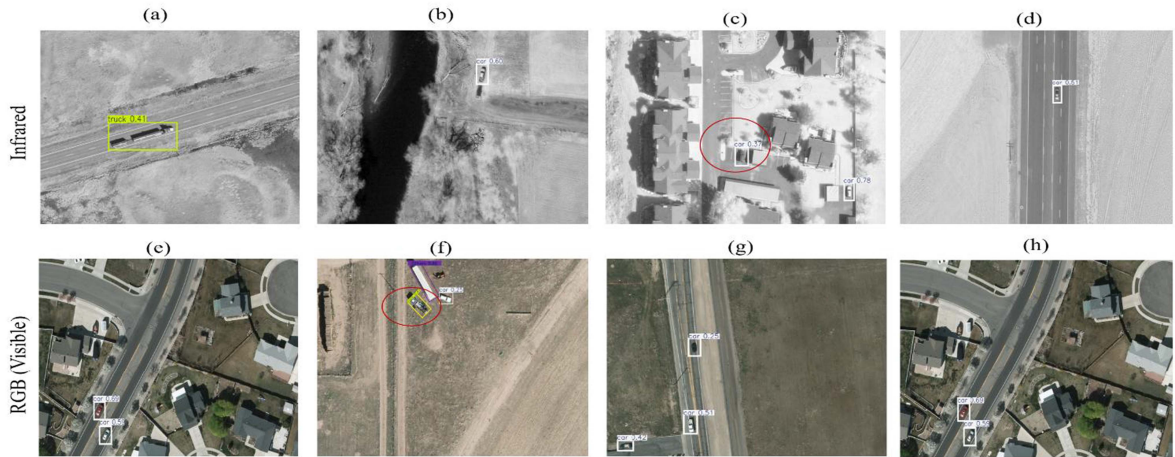


FIGURE 10. Results of object detection in VEDAI dataset by YT-SWIN fusion model in IR, RGB, and Visible Road and large area of grassland environment with tiny object and the red circles represent the failure analysis in detection results.

detecting medium-to-large objects across varied urban and residential landscapes under diverse atmospheric degradations (fog, rain, night), demonstrating the framework’s resilience to severe environmental interference. Conversely, the VEDAI dataset was utilized to assess performance on extremely small objects characterized by complex natural backgrounds (grasslands), significant target distance, and high camouflage, thereby testing the limits of feature discriminability. As quantitatively evidenced by the per-class Average Precision (AP) results presented in Figs. 7 and 9, the model achieves competitive performance across all classes in both datasets maintaining high recall for large vehicles in DroneVehicle and precise localization for diminutive targets in VEDAI which collectively confirms that the proposed adaptive fusion mechanism generalizes effectively to objects of all scales and complexities, ensuring comprehensive detection capability.

F. ABLATION STUDY

Firstly, to justify the YOLOv10n as RGB backbone, we compare it with several YOLO architectures from different generations, all integrated with complete YT-SWIN fusion framework (LSD+SEIM+HCAN) shown in Table 3. The evaluation is conducted on the DroneVehicle dataset to ensure consistent environmental condition.

TABLE 3. Performance Analysis With Yolo Backbones

Backbone	mAP@50	Params	GFLOPs	FPS
YOLOv4-CSP	73.5%	27.6	41.3	29.8
YOLOv5n	75.9%	23.5	39.8	36.4
YOLOv7-Tiny	77.2%	25.2	42.1	33.7
YOLOv8n	76.8%	23.1	38.9	35.2
YOLOv10n (Ours)	78.1%	24.1	41.1	34.5

We evaluate not only the absolute detection performance but also the Robustness Gap (Δ_{rob}). The robustness gap is defined as the performance drop between the Clear subset and the combined Adverse (Fog, Low-light, Night) subset. A smaller gap indicates superior resilience to environmental degradation, isolating the contribution to robustness, whereas an increase in absolute mAP (P_{abs}) indicates an improvement in model capacity. All configurations are built upon the YOLOv10n backbone for RGB feature extraction.

As detailed in Table 4, we evaluate performance by progressively integrating our innovations into a baseline model that uses YOLOv10n with simple concatenation for multimodal fusion. This baseline achieves 69.78% mAP@50. By adding the Learnable Spectral Decomposition (LSD) module improves mAP@50 by 3.4% points to 73.18%, demonstrating the efficacy of adaptive frequency-domain filtering

TABLE 4. Performance Analysis With Different Component Configuration

Model	mAP@50 (P_{abs})	Params (M)	GFLOPs	FPS
A1:YOLOv10n+Concat (Baseline)	69.78%	24.8	38.6	48.2
A2: A1+LSD	73.18%	P25.3	39.1	46.8
A3: A2+SEIM	76.88%	25.7	39.3	45.3
A4: A3+GCM	79.68%	26.2	40.2	43.1
A5: A4+Swin T (Full HCAN)	83.08%	28.3	41.1	32.6
A6: A5+Adaptive channel	82.02%	24.1	38.2	38.5

TABLE 5. Robustness and Capacity Analysis With Different Component Configuration

Model	mAP@50 (P_{abs})	Robustness Gap (Δ_{rob})	Primary Contribution
A1:YOLOv10n+Co ncat (Baseline)	69.78%	22.72%	Capacity
A2: A1+LSD	73.18%	19.32%	Capacity + Minor Robustness
A3: A2+SEIM	76.88%	15.62%	Robustness
A4: A3+GCM	79.68%	12.82%	Capacity
A5: A4+Swin T (Full HCAN)	83.08%	8.82%	Capacity
A6: A5+Adaptive channel	82.02%	9.84%	Efficient

for feature separation. The subsequent addition of the Self-Supervised Environment Inference Module (SEIM) yields a further gain of 3.7 percentage points (76.88% mAP@50), confirming the value of automated environmental context learning. Incorporating the Global Context Module (GCM) enhances performance to 79.68% mAP@50 by capturing long-range dependencies. The complete Hierarchical Cross-Modal Attention Network (HCAN) with Swin Transformer blocks achieves the peak accuracy of 83.08% mAP@50, underscoring the critical role of joint global-local feature modeling and adaptive fusion within the YOLOv10n framework. Finally, applying Adaptive Channel Pruning reduces the parameter count by 15% and computational cost by 17% compared to full HCAN, with only a marginal performance drop to 82.02% mAP@50, establishing an optimal balance between accuracy and efficiency for deployment. This stepwise analysis conclusively validates that each architectural component contributes significantly to enhancing the base YOLOv10n backbone's robustness and effectiveness within the YT-SWIN framework.

Table 5 clearly isolate the functional roles of each component. The introduction of the Self-Supervised Environment Inference Module (SEIM) yields the most significant reduction in the robustness gap (from 19.32% to 15.62%), confirming its primary role in enhancing environmental adaptability. In contrast, the Global Context Module (GCM) and the full HCAN architecture predominantly increase absolute model capacity, boosting P_{abs} by 2.8% and 3.4% respectively, with only marginal further reductions in Δ_{rob} . Interestingly, the Learnable Spectral Decomposition Block (LSD) provides a balanced improvement in both domains. Finally, Adaptive Channel Pruning (A6) restores efficiency without eroding the robustness gained by SEIM. This analysis conclusively

demonstrates that our framework achieves high capacity and high robustness through the synergistic integration of specialized modules.

VI. CONCLUSION

The study presented YT-SWIN, a novel adaptive fusion framework for robust all weather object detection with IR and RGB environment in UAV multimodal remote sensing. The proposed method addresses the critical challenges of unreliable sensor data in adverse conditions. The model introducing a self-supervised environment inference module that autonomously characterizes scene conditions, a learnable spectral decomposition block for adaptive frequency-domain feature separation, and a hierarchical cross-modal attention network built upon Swin Transformer blocks for context-aware fusion. Extensive experiments on DroneVehicle and VEDAI datasets demonstrate that YT-SWIN achieves state-of-the-art performance, outperforming existing methods by significant margin, particularly in foggy, low light, infrared, and RGB visible conditions, while maintaining real-time inference speeds suitable for UAV deployment. This makes it highly suitable for critical drone sensing applications, including urban traffic monitoring and intelligent environmental perceptions. This work provides a practical and effective solution for autonomous remote sensing systems operating in unstructured outdoor scenarios. Future work focus on extending the framework to incorporate additional modalities and further optimizing it for edge deployment.

REFERENCES

- [1] F. Qingyun and W. Zhaokui, "Cross-modality attentive feature fusion for object detection in multispectral remote sensing imagery," *Pattern Recognit.*, vol. 130, 2022, Art. no. 108786, doi: [10.1016/j.patcog.2022.108786](https://doi.org/10.1016/j.patcog.2022.108786).
- [2] M. Sharma et al., "YOLOrs: Object detection in multimodal remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1497–1508, 2021, doi: [10.1109/JSTARS.2020.3041316](https://doi.org/10.1109/JSTARS.2020.3041316).
- [3] J. Xie, J. Nie, B. Ding, M. Yu, and J. Cao, "Cross-modal local calibration and global context modeling network for RGB-infrared remote-sensing object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8933–8942, 2023, doi: [10.1109/JS-TARS.2023.3315544](https://doi.org/10.1109/JS-TARS.2023.3315544).
- [4] M. Yuan and X. Wei, "C²Former: Calibrated and complementary transformer for RGB-infrared object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403712, doi: [10.1109/TGRS.2024.3376819](https://doi.org/10.1109/TGRS.2024.3376819).
- [5] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6700–6713, Oct. 2022, doi: [10.1109/TCSVT.2022.3168279](https://doi.org/10.1109/TCSVT.2022.3168279).
- [6] X. Zhang et al., "Cross-sensor SAR image target detection based on dynamic feature discrimination and center-aware calibration," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5209417, doi: [10.1109/TGRS.2025.3559618](https://doi.org/10.1109/TGRS.2025.3559618).
- [7] Z. Sun et al., "Arbitrary-direction SAR ship detection method for multiscale imbalance," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5208921, doi: [10.1109/TGRS.2025.3559701](https://doi.org/10.1109/TGRS.2025.3559701).
- [8] C. Zheng et al., "Cross-modal progressive perspective matching network for remote sensing image-text retrieval," *IEEE Trans. Multimedia*, vol. 27, pp. 3966–3978, 2025, doi: [10.1109/TMM.2025.3535365](https://doi.org/10.1109/TMM.2025.3535365).
- [9] J. Nie, X. Wang, R. Hou, G. Li, H. Chen, and W. Zhu, "Dynamic spatio-temporal graph reasoning for VideoQA with self-supervised event recognition," *IEEE Trans. Image Process.*, vol. 33, pp. 4145–4158, 2024, doi: [10.1109/TIP.2024.3411448](https://doi.org/10.1109/TIP.2024.3411448).

- [10] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017, doi: [10.1109/TPAMI.2016.2577031](#).
- [11] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 2844–2853.
- [12] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021, doi: [10.1109/TPAMI.2020.2974745](#).
- [13] J. Han, J. Ding, N. Xue, and G. S. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 2786–2795.
- [14] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2980–2988.
- [15] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511, doi: [10.1109/TGRS.2021.3062048](#).
- [16] X. Yang, J. Yan, and Z. Feng, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 3163–3171, doi: [10.1609/aaai.v35i4.16426](#).
- [17] Q. Liu, R. Liu, B. Zheng, H. Wang, and Y. Fu, "Infrared small target detection with scale and location sensitivity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 17490–17499.
- [18] Y. Dai, Y. Wu, F. Zhou, and K. Barnard, "Asymmetric contextual modulation for infrared small target detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2021, pp. 950–959.
- [19] B. Li et al., "Dense nested attention network for infrared small target detection," *IEEE Trans. Image Process.*, vol. 32, pp. 1745–1758, 2023, doi: [10.1109/TIP.2022.3199107](#).
- [20] Y. Dai, Y. Wu, F. Zhou, and K. Barnard, "Attentional local contrast networks for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9813–9824, Nov. 2021, doi: [10.1109/TGRS.2020.3044958](#).
- [21] M. S. Pillai, G. Chaudhary, and M. Khari, "Real-time image enhancement for an automatic automobile accident detection through CCTV using deep learning," *Soft Comput.*, vol. 25, no. 18, pp. 11929–11940, 2021, doi: [10.1007/s00500-021-05576-w](#).
- [22] J. Terven, D. M. Córdova-Esparza, and J. A. Romero-González, "A comprehensive review of YOLO architectures in computer vision: From YOLOv1 to YOLOv8 and YOLO-NAS," *Mach. Learn. Knowl. Extrac.*, vol. 5, no. 4, pp. 1680–1716, 2023, doi: [10.3390/make5040083](#).
- [23] T. Li and G. Li, "Road defect identification and location method based on an improved ML-YOLO algorithm," *Sensors*, vol. 24, no. 21, 2024, Art. no. 6783, doi: [10.3390/s24216783](#).
- [24] A. Vijayakumar and S. Vairavasundaram, "YOLO-based object detection models: A review and its applications," *Multimedia Tools Appl.*, vol. 83, no. 35, pp. 83535–83574, 2024, doi: [10.1007/s11042-024-18872-y](#).
- [25] M. U. Rehman, H. Eesaar, and Z. Abbas, "Advanced drone-based weed detection using feature-enriched deep learning approach," *Knowl.-Based Syst.*, vol. 305, 2024, Art. no. 112655, doi: [10.1016/j.knosys.2024.112655](#).
- [26] C. Chen et al., "Weakly misalignment-free adaptive feature alignment for UAVs-based multimodal object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 26826–26835.
- [27] X. He, C. Tang, and X. Zou, "Multispectral object detection via cross-modal conflict-aware learning," in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 1465–1474, doi: [10.1145/3581783.3612651](#).
- [28] T. Zhao, M. Yuan, F. Jiang, N. Wang, and X. Wei, "Removal then selection: A coarse-to-fine fusion perspective for RGB-infrared object detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 27, no. 2, pp. 2504–2519, Feb. 2026, doi: [10.1109/TITS.2025.3638627](#).
- [29] X. Sun, Y. Zhu, and H. Huang, "Specificity-guided cross-modal feature reconstruction for RGB-infrared object detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 1, pp. 950–961, Jan. 2025, doi: [10.1109/TITS.2024.3495028](#).
- [30] X. Sun, Y. Yu, and Q. Cheng, "Low-rank multimodal remote sensing object detection with frequency filtering experts," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5637114, doi: [10.1109/TGRS.2024.3446814](#).
- [31] Z. An, C. Liu, and Y. Han, "Effectiveness guided cross-modal information sharing for aligned RGB-T object detection," *IEEE Signal Process. Lett.*, vol. 29, pp. 2562–2566, 2022, doi: [10.1109/LSP.2022.3229571](#).
- [32] J. Wang et al., "Multimodal object detection of UAV remote sensing based on joint representation optimization and specific information enhancement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 12364–12373, 2024, doi: [10.1109/JSTARS.2024.3373816](#).
- [33] A. N. M. Nusari, I. Y. Ozbek, and E. A. Oral, "Automatic vehicle accident detection and classification from images: A comparison of YOLOv9 and YOLO-NAS algorithms," in *Proc. 32nd Signal Process. Commun. Appl. Conf.*, 2024, pp. 1–4, doi: [10.1109/SIU61531.2024.10600761](#).
- [34] O. Byzkrovnyi, K. Smelyakov, and A. Chupryna, "Comparison of potential road accident detection algorithms for modern machine vision system," in *Proc. Int. Sci. Practical Conf.*, 2023, pp. 50–55, doi: [10.17770/etr2023vol3.7299](#).
- [35] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.
- [36] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2021, *arXiv:2010.11929*.
- [37] D. Zhang, H. Zhang, and J. Tang, "Feature pyramid transformer," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 323–339, doi: [10.1007/978-3-030-58604-1_20](#).
- [38] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 10012–10022.
- [39] Y. Xu, Y. Yang, and L. Zhang, "DeMT: Deformable mixer transformer for multi-task learning of dense prediction," *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 3, pp. 3072–3080, 2023.
- [40] N. Carion, F. Massa, and G. Synnaeve, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 213–229, doi: [10.1007/978-3-030-58452-8_13](#).
- [41] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.
- [42] M. Zheng, P. Gao, and R. Zhang, "End-to-end object detection with adaptive clustering transformer," 2020, *arXiv:2011.09315*.
- [43] Z. Dai, B. Cai, Y. Lin, and J. Chen, "UP-DETR: Unsupervised pre-training for object detection with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 1601–1610.
- [44] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6700–6713, 2022. [Online]. Available: <https://www.kaggle.com/datasets/redzapdos123/dronevehicledataset-yolov11-obv>
- [45] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," *J. Visual Commun. Image Representation*, vol. 34, pp. 187–203, 2016. [Online]. Available: <https://www.kaggle.com/datasets/penjin/vedai1024>
- [46] M. Yuan, Y. Wang, and X. Wei, "Translation, scale and rotation: Cross-modal alignment meets RGB-infrared vehicle detection," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 509–525, doi: [10.1007/978-3-031-20077-9_30](#).
- [47] M. Zhou et al., "DMM: Disparity-guided multispectral mamba for oriented object detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, 2025.
- [48] A. Bochkovskiy, C. Y. Wang, and H. Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.
- [49] R. Varghese and S. M., "YOLOv8: A novel object detection algorithm with enhanced performance and robustness," in *Proc. Int. Conf. Adv. Data Eng. Intell. Comput. Syst.*, 2024, pp. 1–6, doi: [10.1109/ADICS58448.2024.10533619](#).
- [50] M. Sharma et al., "YOLOrs: Object detection in multimodal remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1497–1508, 2021, doi: [10.1109/JSTARS.2020.3041316](#).
- [51] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605415, doi: [10.1109/TGRS.2023.3258666](#).
- [52] W. Li, A. Li, X. Kong, Y. Zhang, and Z. Li, "MF-YOLO: Multimodal fusion for remote sensing object detection based on YOLOv5s," in *Proc. 27th Int. Conf. Comput. Supported Cooperative Work Des.*, 2024, pp. 897–903, doi: [10.1109/CSCWD61410.2024.10580205](#).